

Weight — Usability Evaluation

A one-person moderated usability test of the deployed calculator. Everyday Tools — Fordham Wong

Tool fwong29.github.io/reusable-studio-engine/everyday-tools/ (deployed on GitHub Pages)
Method tester given only the link and the scenario below; observed silently, no interface explanation

TESTER

Teen (13–17), highly comfortable with technology and a frequent doer of quick everyday math — the kind of user who enters numbers fast and without thinking. This fits the tool's intended audience: someone whose habit of frictionless quantifying is exactly what the tool tries to interrupt, so his reactions are a fair read on whether the friction lands or just annoys.

TESTING SCENARIO

The tester was given the situation, not the interface:

"You need to add up a few numbers — the kind of thing you'd normally punch into a calculator without a second thought, like totaling an amount or adding a short list. Open this link and work out a couple of calculations however you'd normally do it."

He was encouraged to think out loud, and was not guided unless completely stuck.

FEEDBACK SUMMARY (GOOGLE FORM, 1 RESPONSE)

3/5

overall rating

What is the tool trying to help you do?	"Be mindful of my calculations and how accessible knowledge is in these times." — he read the intent correctly: mindfulness about quantities, not raw speed.
Clearest part of the interface?	"The numbers." — the tap-to-build number key and the tally read cleanly.
Most confusing part?	"The key combos like shift + space." — the keyboard controls were not discoverable.
How did it feel when it intervened?	"Annoying, as though it was too inefficient." — he felt the friction, but experienced it as an obstacle rather than a pause.
Would you realistically use it?	"No."
Design improvement?	"Make the instructions more clear."
Functionality improvement?	"Make it so that I can do more math things."

OBSERVED BEHAVIOR

- **Immediate success:** tapping the number key to build a value — he understood the tap-per-unit rule without being told.
- **Hesitation:** at the keyboard; he tried number-row keys expecting them to type digits, and the Shift/space mapping was invisible until discovered by accident.
- **Friction read as inefficiency:** the deliberate cost landed emotionally, but with no framing he read it as the tool being slow rather than as an intentional pause.

REFLECTION — WHAT I WILL CHANGE

Two clear signals, and I'm taking both:

1. Make the instructions clearer. The single caption isn't enough. I'll add a short, always-visible legend for the controls — what the number key and operator key do, and the keyboard shortcuts (space = one tap, Shift = cycle operator, = to solve) — so the confusion he hit at "shift + space" disappears. This is the important distinction the test surfaced: the tap cost is *intended* friction and I'm keeping it, but the confusion over how to drive the interface is *accidental* friction, and that's what made the whole thing feel "inefficient." Clear guidance should let the deliberate pause read as a pause, not as "I don't know how to use this."

2. Let it do more math. He wanted more capability, and a tool that only adds and subtracts gives the friction little to be about. I'll widen what it can compute (more operations, cleaner handling of longer expressions) while keeping the one rule that defines it — numbers still cost taps. More range, same ritual.

What I won't change is the core cost itself. A tester wanting it faster is expected for a tool built to slow one specific moment down; the fix is not to remove the friction but to make it legible and give it more worth doing.